

MATHAN P | Junior Data Scientist

Spice Garden Layout,
Marathahalli,
Bengaluru, Karnataka - 560037

+91 7339404085

mathanponraj03@gmail.com

PROFESSIONAL SUMMARY

A highly motivated and detail-oriented **B.E. Computer Science graduate** with hands-on experience in **Python, Java, SQL, data analysis, preprocessing, and machine learning**. Possesses strong technical aptitude with practical experience in **EDA, feature engineering, model evaluation, ETL pipelines, and ML deployment**, demonstrated through real-world academic and internship projects. Developed and deployed an **XGBoost laptop price prediction and recommendation engine** using Streamlit, achieving **~0.91 R²**, and designed a YouTube ETL pipeline processing **50+ trending videos per run** using Python, Pandas, SQL, and Apache Airflow. Led a **4-member team** in the Naan Mudhalvan × IBM collaborative program to successfully deploy a Gold Commodity News Sentiment Analysis project using Flask. Strong analytical, communication, and collaboration skills with the ability to learn quickly and work effectively in fast-paced team environments, seeking an **Entry-Level Data Scientist** opportunity.

EDUCATION

GOVERNMENT COLLEGE OF ENGINEERING, BODINAYAKKANUR

Undergraduate Degree: B.E in Computer Science and Engineering

(2020 - 2024)

Coursework: Data Structures & Algorithms, Software Development, Computer Networks, DBMS, Operating Systems

TECHNICAL EXPERTISE

- **Programming & Querying:** Python, SQL, Java
- **Data Science:** Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Data Preprocessing
- **Machine Learning:** Regression, Classification, Clustering, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Model Deployment
- **Deep Learning:** TensorFlow, Keras, PyTorch, Neural Networks, CNN, RNN, LSTM
- **Data Engineering:** ETL Pipelines, Apache Airflow, SQLite
- **Deployment & Visualization:** Streamlit, Flask, Power BI, Excel, Matplotlib, Seaborn
- **Soft Skills:** Problem Solving, Communication, Collaboration, Analytical Thinking

INTERNSHIP EXPERIENCE

Data Science Intern – RINEX

(Mar 2025 - May 2025)

- Developed an end-to-end **laptop price prediction and recommendation system** using Python, Pandas, Scikit-learn, and XGBoost to analyze hardware specifications and predict laptop prices.
- Performed **data cleaning and preprocessing**, including missing-value handling, duplicate removal, categorical encoding, and feature standardization to prepare reliable datasets for model training.
- Trained and evaluated an **XGBoost Regressor**, achieving an **~0.91 R² score** on unseen data and validating model performance using regression evaluation metrics.
- Deployed the model through **Streamlit** and implemented budget-based filtering and ranking logic to generate **Top 5 laptop recommendations**, providing users with an interactive way to compare suitable options.

Professional Employability Intern – CDAC, Chennai

(Jul 2023 - Aug 2023)

- Developed a **Python-based security authentication application** integrated with sensors to

monitor unauthorized door access and trigger real-time security alerts.

- Implemented an automated alert mechanism to send **access notifications through Telegram and WhatsApp**, enabling faster awareness of potential unauthorized entry.
- Gained hands-on exposure to **Python programming, sensor integration, cloud computing concepts, application troubleshooting, and technical problem-solving** through the internship project.

PROJECTS

YouTube Trending Analytics & Automated ETL Pipeline

Python | Pandas | SQL | SQLite | Apache Airflow | YouTube Data API v3

Goal: Automate the collection and analysis of trending YouTube data to identify video performance, audience engagement, and channel trends.

- **Designed** an automated ETL pipeline using the **YouTube Data API v3** to extract **50+ trending videos per run**, providing structured data for downstream analytics.
- **Implemented** data cleaning and transformation with Python and Pandas, including missing-value handling, duplicate removal, data-type conversion, and required-field validation.
- **Developed** a SQLite data layer and performed **SQL-based EDA** using filtering, aggregation, sorting, and engagement metrics to identify top-performing videos and trending channels.
- **Orchestrated** the complete workflow using **Apache Airflow DAGs**, implementing scheduling, task dependencies, retry handling, and execution monitoring to detect and troubleshoot failed tasks.
- **Result:** Established a repeatable, monitored analytics workflow that transformed raw API responses into **analysis-ready datasets** for efficient SQL retrieval and content-performance analysis.

Laptop Price Prediction & Intelligent Recommendation System

Python | Pandas | NumPy | Scikit-learn | XGBoost | Streamlit

Goal: Predict laptop prices from hardware specifications and recommend suitable high-value laptops based on user budget and preferences.

- **Engineered** the dataset by performing data cleaning, missing-value handling, duplicate removal, categorical encoding, and feature preparation to create an analysis-ready dataset.
- **Performed** feature engineering and exploratory analysis to identify relationships between laptop specifications and pricing, supporting model development and selection.
- **Developed and evaluated** an **XGBoost Regressor**, achieving an **R² score of ~0.91** in predicting laptop prices from hardware specifications.
- **Implemented** a recommendation engine that applies user budget constraints and predicted prices to rank and generate **Top 5 laptop recommendations**.
- **Deployed** the complete solution as an interactive **Streamlit application**, enabling users to enter preferences, view predictions, and compare recommended laptops.

Gold Commodity News Sentiment Analysis

Python | NLP | Scikit-learn | TF-IDF | Flask | Machine Learning

Goal: Analyze sentiment from gold commodity news and historical data to identify positive, negative, and neutral market sentiment.

- **Led a 4-member team** under the **Naan Mudhalvan × IBM collaborative program** to develop an NLP-based sentiment analysis solution using gold commodity news and historical data.
- **Collected and prepared** data from **news sources and historical datasets**, applying text cleaning, preprocessing, and data quality checks to create a reliable dataset for analysis.
- **Implemented TF-IDF feature extraction and machine learning classification** to convert news text into numerical features and predict sentiment categories.

- **Evaluated** the model using classification metrics and achieved **91.65% accuracy**, demonstrating effective sentiment classification on the prepared dataset.
- **Deployed** the solution using **Flask**, enabling users to submit news content and obtain sentiment predictions through an interactive application.

CAREER OBJECTIVE

A motivated and analytical **Computer Science graduate** with strong problem-solving and programming skills, supported by a solid foundation in **Python, Java, SQL, and Data Science**. Possess strong analytical abilities and practical exposure to machine learning and data-driven projects, making me well-suited for a **Junior/Entry-Level Data Scientist** role. Also familiar with **Linux, troubleshooting, and technical support**, with the flexibility to contribute to entry-level **Software, Data Engineering Support, or Technical Support** roles.

CERTIFICATIONS

- Data Science and Machine Learning from RINEX
- Advanced EXCEL and SQL for Data Analytics from Rinex
- Python Programming from RINEX
- Data Analytics Virtual Internship - Deloitte
- AWS (pursuing) from Rinex

PERSONAL DETAILS

1. Linked in - [mathan03](#)
2. GitHub Repository - [mathan-ponraj](#)
3. Phone Number: +91 7339404085
4. Email: mathanponraj03@gmail.com

I **MATHAN P**, do hereby confirm that the information provided above is true to the best of my knowledge.

SIGNATURE: _____

(MATHAN P)